

Sigma and pi bonds (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Describe sigma and pi bonds in terms of atomic orbital overlap.
- Deduce the presence of sigma bonds and pi bonds in organic and inorganic molecular compounds and polyatomic ions.

Section S2.2.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>) described how covalent bonds are formed when single electrons from different atoms form a pair. This pair is then shared between the atoms: it is the electrostatic attraction of nuclei in those atoms for the shared pair of electrons that constitutes the covalent bond.

Section S1.3.3–4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>) covered the shapes of orbitals. The s-orbitals were spherical and p-orbitals had a dumb-bell shape oriented at 90° to each other. How is it possible that combinations of these could produce bond angles such as 109.5° found in a tetrahedral molecular geometry?

The next section goes into more depth as what is actually happening when electrons, located in different orbitals, form a pair.

Sigma bonds

Electrons are found in orbitals; s, p, d and f (see [section S1.3.3–4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>)). When the main energy level is filled with the maximum number of electrons it can hold, the octet rule has been fulfilled and atoms are stable. When an orbital is partially empty, it will gain or lose electrons (ionic bond), or it will share electrons (covalent bond) so it can then fulfil the octet rule.

Considered what happens to the different orbitals in an atom when the electrons they contain form a pair. You can see this in **Figure 1**, which shows a single bond forming between s-orbitals in two hydrogen atoms and results in a hydrogen molecule.

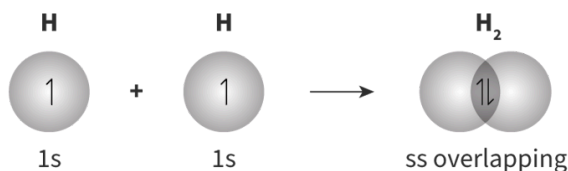


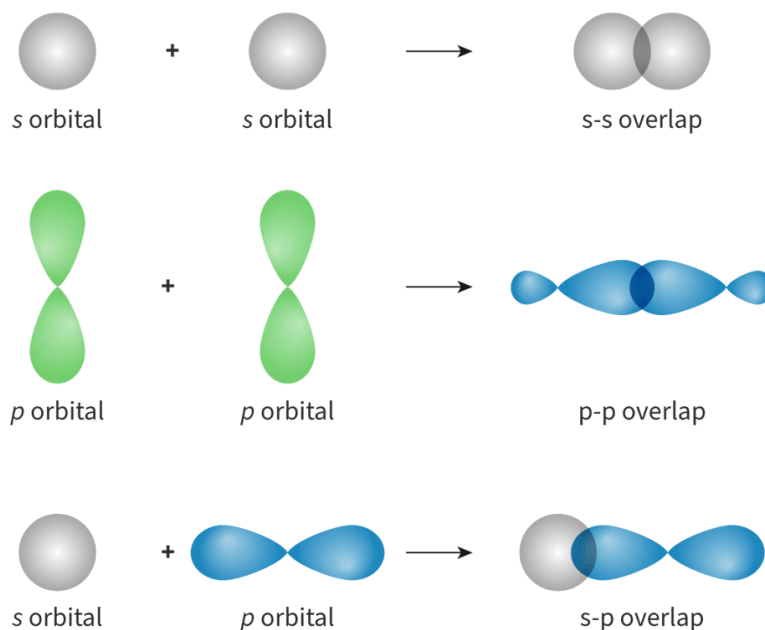
Figure 1. The head-on overlap of two s-orbitals in hydrogen. More information for figure 1

The diagram illustrates the formation of a hydrogen molecule through the head-on overlap of two s-orbitals. On the left, there are two hydrogen atoms represented by circles, each labeled with "H" and showing "1" inside to indicate the single electron in the 1s orbital. The two atoms are positioned separately with a plus sign between them. An arrow points to the right, leading to a depiction of the two overlapping s-orbitals, which is represented by two overlapping circles labeled "H₂." The region where the circles overlap is shaded and labeled "ss overlapping," indicating the shared space where the electrons form a sigma bond.

[Generated by AI]

As you can see in **Figure 1**, a covalent bond is formed when two orbitals overlap, each one containing one electron. If this orbital-overlap is a head-on overlap, it is known as a sigma bond, and the electron pair is likely to be found in that overlapping shared space. Single bonds are always sigma bonds, which are represented with the Greek symbol for sigma, σ .

Sigma bonds can be formed from the head-on overlap of s-orbitals to s-orbitals, s-orbitals to p-orbitals, or p-orbitals to p-orbitals (**Figure 2**). Note that this head-on overlap can also be referred to as an axial overlap or end-to-end overlap.

**Figure 2.** Sigma bonds are formed by the head-on overlap of s- and p-orbitals. More information for figure 2

This diagram illustrates the formation of sigma bonds through the overlap of atomic orbitals. It consists of three main sections, each demonstrating a different type of orbital overlap.

1. s-s Overlap:

2. Two s orbitals, depicted as gray circles, are shown side by side. An arrow points to the right, indicating the direction of overlap, leading to the formation of an s-s overlap. The s-s overlap is represented by two gray circles overlapping each other.

3. p-p Overlap:

4. Two p orbitals, illustrated as green, shaped like dumbbells, are next to each other. An arrow points to their overlap, creating a p-p overlap. The result is shown with two blue dumbbell shapes overlapping in the central region.

5. s-p Overlap:

6. An s orbital, depicted as a gray circle, is adjacent to a p orbital, shown as a blue dumbbell shape. An arrow indicates the overlap direction, resulting in an s-p overlap. The overlap combines the gray circle (s orbital) and one lobe of the blue dumbbell (p orbital) merging together.

Each part of the diagram highlights the direct head-on overlap characteristic of sigma bonds.

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An orbital does not indicate the exact location of an electron, but rather represents where an electron is most likely to be found (see [section S1.3.3–4](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>)). This area of 'overlap' is based on the same principle: it indicates the area where the probability of finding an electron is highest.

Pi bonds

Molecules and compounds can also contain double or triple bonds (see [section S2.2.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/>)). Knowing that electron pairs are negatively charged and repel other electron pairs, where do you think this second or third pair of electrons can fit between the two bonded atoms after the sigma bond has formed?

The electron pair forming the second or third bond cannot lie in the same axial plane as the electron pair forming the sigma bond as they would repel each other: they must lie somewhere else between the two atoms.

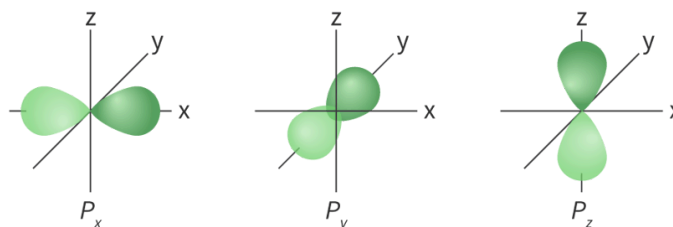


Figure 3. p-orbitals exist on the x-, y- and z-axes.

 More information for figure 3

This image shows three diagrams of p-orbitals positioned along different axes. The first diagram on the left depicts a p-orbital on the x-axis, labeled as P_x , with two lobes extending horizontally across the axis. The middle diagram

shows a p-orbital on the y-axis, labeled as P_y , also displaying two lobes extending across the horizontal axis. The third diagram on the right illustrates a p-orbital on the z-axis, labeled as P_z , with two lobes vertically aligned. Each orbital has a dumbbell shape, which is characteristic of p-orbitals, and the three diagrams signify the orientation of these orbitals along the x, y, and z axes respectively.

[Generated by AI]

To form a double or triple bond while keeping the sigma bonded electrons far enough away, only p-orbitals are used. Whereas s-orbitals have a spherical shape, p-orbitals have a dumb-bell shape (see [S1.3.3-4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>)). As you can see in **Figure 3**, separate dumb-bell shaped, p-orbitals are found on the x-, y- and z-axes.

Try to visualise these orbitals overlapping. The two p_x orbitals can easily overlap head-on to form a sigma bond, but the parallel p_y and p_z orbitals cannot; they can only overlap sideways (**Figure 4**).

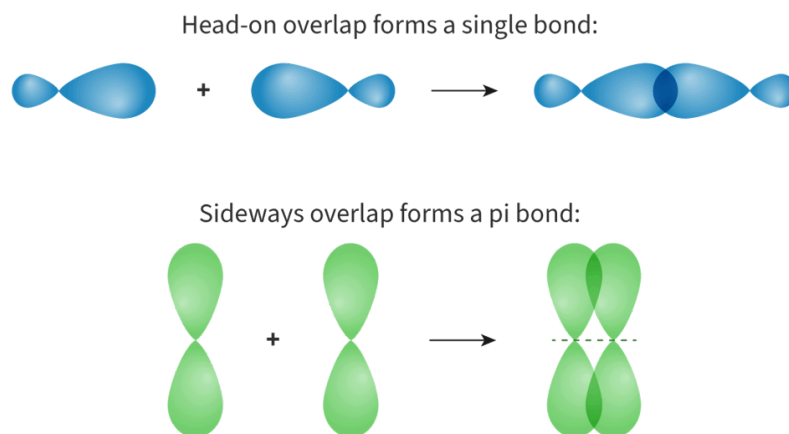


Figure 4. Two p-orbitals can overlap head on while p_y and p_z overlap sideways.

 More information for figure 4

The image illustrates how p-orbitals combine to form different types of chemical bonds. The top section shows a head-on overlap of two blue p-orbitals, labeled as "Head-on overlap forms a single bond," leading to a sigma bond. The bottom section demonstrates a sideways overlap of two green p-orbitals, labeled as "Sideways overlap forms a pi bond," resulting in a pi bond. The overlapping areas indicate where the bonds are formed.

[Generated by AI]

If the orbital overlap is a sideways or lateral overlap, it is known as a pi bond and is represented with the Greek symbol for pi, π (**Figure 5**).

Pi bonds are found in double and triple bonds and have their electron density concentrated above and below the bond axis (**Figure 6**).

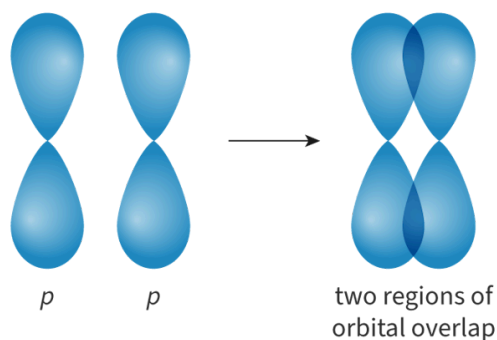


Figure 5. The sideways overlap of parallel p-orbitals.

More information for figure 5

The image is a diagram illustrating the sideways overlap of parallel p-orbitals leading to the formation of pi bonds. On the left, there are two separate p-orbitals depicted as two sets of blue teardrop shapes adjacent to each other. An arrow points towards the right, indicating the process of interaction. On the right, the p-orbitals overlap, forming a pair of joined blue teardrop shapes. The diagram is labeled with 'p' beneath each isolated p-orbital and 'two regions of orbital overlap' beneath the combined shape, showing the regions where electron density increases due to sideways overlap, characteristic of pi bonding.

[Generated by AI]

The types of bonds are summarised in **Table 1**.

Table 1. Types of bonds.

Bond	Pairs of electrons shared	Number of bonds
Single	1	1 sigma (σ)
Double	2	1 sigma (σ) and 1 pi (π)
Triple	3	1 sigma (σ) and 2 pi (π)

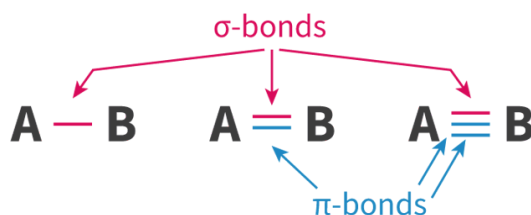


Figure 6. Sigma and pi bonds in single, double and triple bonds.

More information for figure 6

The diagram shows the types of bonds in relation to single, double, and triple bonds between atoms labeled as 'A' and 'B'.

1. Single Bond (A-B):

2. Represented by a single line between A and B.

3. Associated with a sigma (σ) bond, indicated by an arrow pointing at the single bond with the label ' σ -bonds'.

4. Double Bond (A=B):

5. Represented by a double line between A and B.

6. Consists of one sigma (σ) bond and one pi (π) bond.

7. Arrows labeled ' σ -bonds' and ' π -bonds' point toward the double bond.

8. Triple Bond (A $\equiv$ B):

9. Represented by a triple line between A and B.

10. Contains one sigma (σ) bond and two pi (π) bonds.

11. Arrows labeled ' σ -bonds' and ' π -bonds' point to the corresponding sections of the triple bond.

The diagram effectively distinguishes between these bond types and how sigma and pi bonds are distributed in each case.

[Generated by AI]

Bond rotation is possible in a sigma bond as there is only one region of electron density, located along the inter-nuclear axis. As a pi bond has two regions of electron density, located above and below the plane of the inter-nuclear axis, it is not possible for a pi bond to rotate.

Worked example 1

Identify the number of sigma and pi bonds in a molecule of but-2-ene-1,4-dioic acid (**Figure 7**).

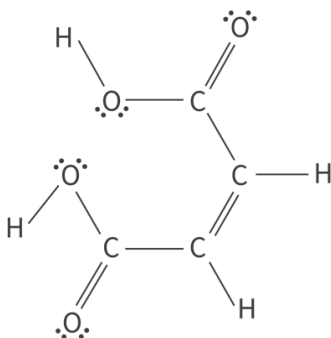


Figure 7. But-2-ene-1,4-dioic acid.

 More information for figure 7

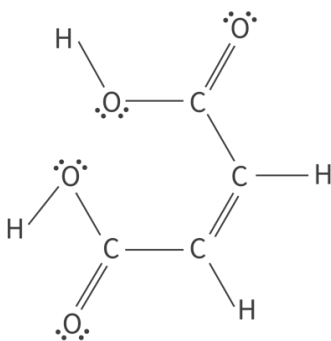
The image depicts the molecular structure of but-2-ene-1,4-dioic acid. The chemical structure presents a chain of four carbon atoms with various bonds: single and double bonds depicted as lines connecting the atoms.

From left to right, the first carbon (C) atom is attached to a hydroxyl group (OH) and an oxygen double-bonded (O) above; the second carbon is linked to a hydrogen (H) atom and has a double bond with the third carbon atom. Both the third and fourth carbon atoms have single bonds with hydrogen atoms, and the fourth carbon is double-bonded to an oxygen atom and single bonded to another oxygen atom, the latter bonded to a hydrogen atom, completing the carboxyl group.

The structure shows the characteristic single and double bonds: sigma bonds depicted by lines between atoms and pi bonds depicted by double lines, illustrating the molecule's potential reactivity and geometry. The image serves to illustrate the chemical composition and bonding of the molecule.

[Generated by AI]

First you must identify the number of single, double and triple bonds.



There are 8 single bonds, each one is a sigma bond.

There are 3 double bonds, each one contains one sigma and one pi bond.

Therefore, there are 11 sigma and 3 pi bonds in the molecule above.

Worked example 2

Identify the number of sigma and pi bonds in a cyanide ion, CN^- .

First you must draw the Lewis formula so that you can identify the number of single, double and triple bonds.

Lewis formula:



There are 0 single bonds and 1 triple bond.

Each triple bond contains one sigma and two pi bonds.

Therefore, there are a total of 1 sigma and 2 pi bonds in the ion above.

Study skills

Even though a 2D representation of a pi bond will look like it has two distinct areas of electron density, in reality the electrons being shared can be found anywhere within those two 'lobes' at any time.

It is important to acknowledge that for there to be a pi bond (or two), there is first a sigma bond between the two atoms. Drawing this is difficult because there are so many particles occupying such a close space. Alternate notations for drawing are often used in order to show more information, for example in the diagram of ethene, C_2H_4 , in **Figure 8**.

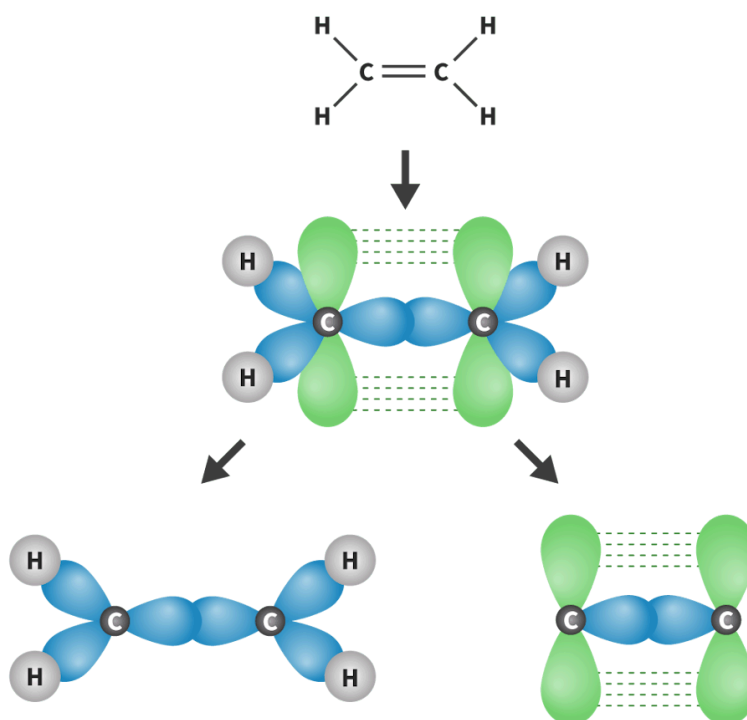


Figure 8. Representing multiple bonds together can be tricky, as ethene shows.

 More information for figure 8

The image is a diagram illustrating the molecular structure of ethene (C_2H_4). At the top, the structural formula $H_2C=CH_2$ is shown, representing ethene. Below it, there are three representations:

- 1. Central Image:** The central diagram shows the carbon (C) and hydrogen (H) atoms with overlapping oval shapes representing orbitals. The p-orbitals from each carbon atom are drawn separate and are connected via dashed lines to signify a pi bond, pi (π), above and below a horizontal sigma bond between the two carbon atoms.
- 2. Left Image:** This diagram depicts the sigma bonds alone between the carbon and hydrogen atoms, clearly showing the sigma bond between the two carbon atoms without the pi bond.

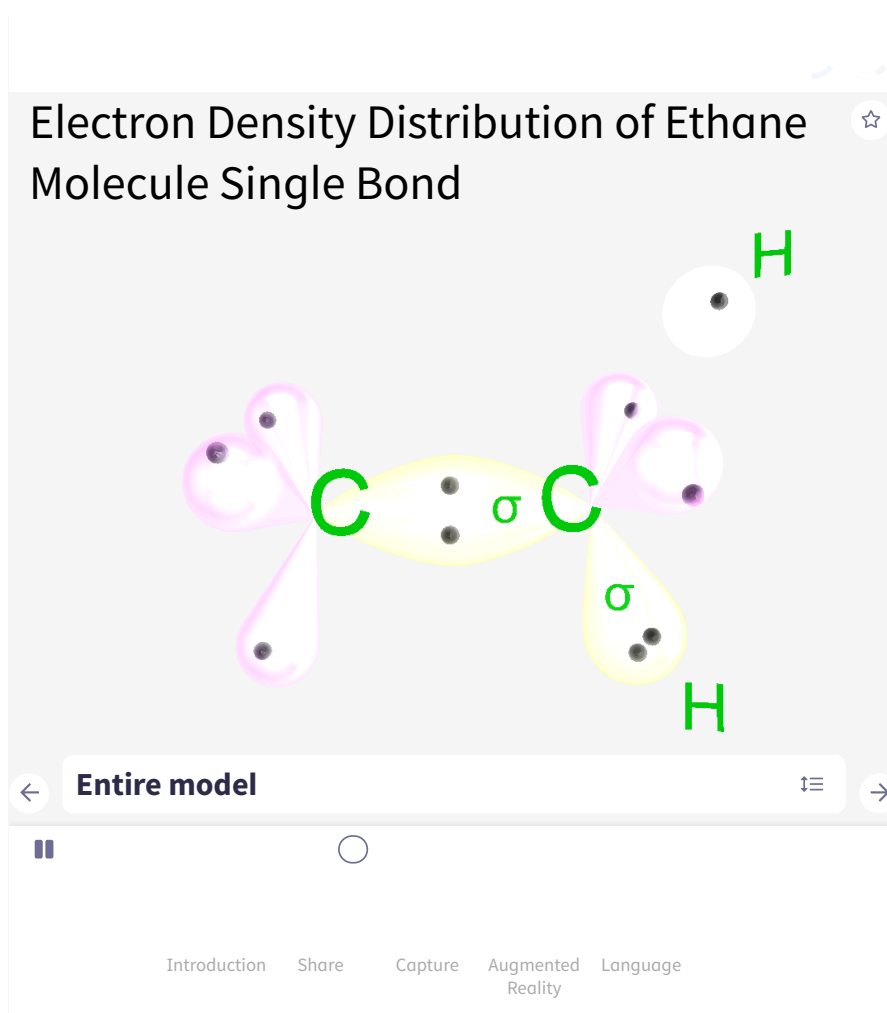
3. **Right Image:** This shows only the pi bonds, highlighting the overlapping nature of the p-orbitals and the absence of the sigma bond in this particular view.

The diagram conveys how diagrams are used to represent complex chemical bonding by isolating sigma and pi bonds visually for clarity, even though this does not show physical reality where the p-orbitals would be overlapping.

[Generated by AI]

As you can see in **Figure 8**, the p-orbitals in the pi bond are drawn apart from each other with lines connecting the p-orbitals and the symbol for pi, π , is used. Although the p-orbitals are actually overlapping, they are drawn this way to more clearly show the presence of the sigma bond, even though this diagram is not an accurate representation of the overlapping p-orbitals

Using a 3D simulator to visualise this can be more effective than a 2D diagram. Use the models following the links to help you visualise the orbital overlap in three organic molecules: ethane, ethene and ethyne, which have carbon-carbon single, double and triple bonds, respectively.



Interactive 1. Electron Density Distribution of Ethane Molecule Single Bond.

More information for interactive 1

An interactive three-dimensional illustration exhibits the electron density distribution of the ethane molecule's single bond. It displays two sp^3 -hybridized carbon atoms, each with four lobes forming a tetrahedral

geometry. Each lobe contains one electron. These two carbon atoms interact, with one lobe from each carbon overlapping to form a sigma (σ) bond between C–C.

Three hydrogen atoms, each with an s orbital containing one electron, interact with the remaining three orbitals of each carbon atom, forming sigma bonds (σ) between C–H.

A carousel selector with arrows is at the bottom, displaying the entire model. Users can rotate the model to view it from different angles. Below it is a seek bar with a play or pause button, allowing users to demonstrate the steps of bond formation.

At the bottom, introduction, share, capture, augmented reality, and language options are available.

The introduction option displays the following details:

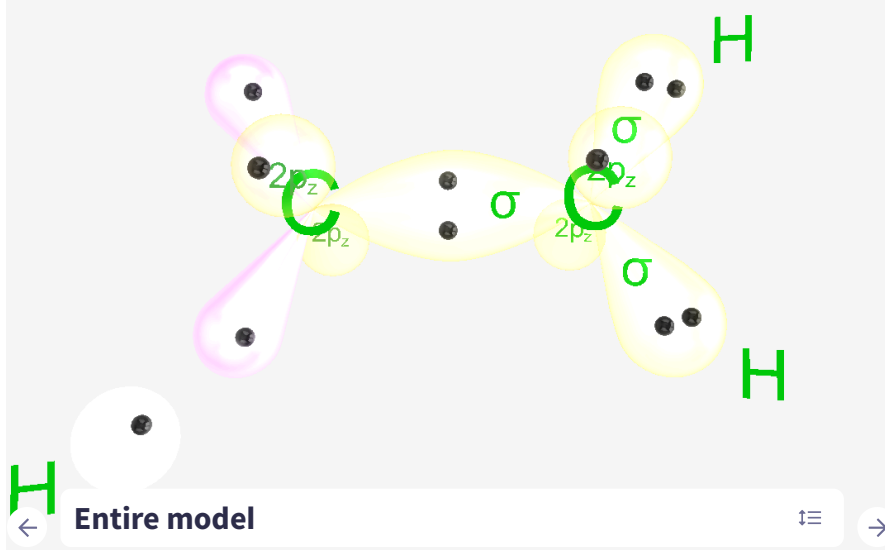
The sp^3 hybridization is used to describe the bonding in the ethane molecule.

When a bond is formed between carbon atoms, two sp^3 orbitals overlap to form a σ -bond. This results in a simple bond, consisting of one σ -bond.

When a bond is formed between a carbon atom and a hydrogen atom, the sp^3 orbital and the s-orbital overlap to form a σ -bond, resulting in a simple bond.

This animation helps users understand the hybridization of carbon, the formation of sigma bonds in ethane, and electron density distribution.

Electron Density Distribution of Ethene Molecule Double Bond



Introduction Share Capture Augmented Reality Language

Interactive 2. Electron Density Distribution of Ethene Molecule Double Bond.

More information for interactive 2

An interactive animation illustrates the electron density distribution of the ethene molecule's double bond. It displays two sp^2 -hybridized carbon atoms in a trigonal planar geometry. Each sp^2 carbon has three sp^2 lobes in the trigonal plane and a $2p_z$ -unhybridized lobe perpendicular to the plane. Each lobe contains one electron. These two carbon atoms interact, with one sp^2 lobe from each carbon overlapping to form a sigma (σ) bond between C-C.

Two hydrogen atoms, each with an s-orbital containing one electron, interact with the remaining two sp^2 orbitals of each carbon atom, forming sigma (σ) bonds between C-H.

The unhybridized $2p_z$ orbitals on each carbon overlap sideways, forming a pi (π) bond, resulting in a C = C double bond.

A carousel selector with arrows is at the bottom, displaying the entire model. Users can rotate the model to view it from different angles. Below it is a seek bar with a play or pause button, allowing users to demonstrate the steps of bond formation.

At the bottom, introduction, share, capture, augmented reality, and language options are available.

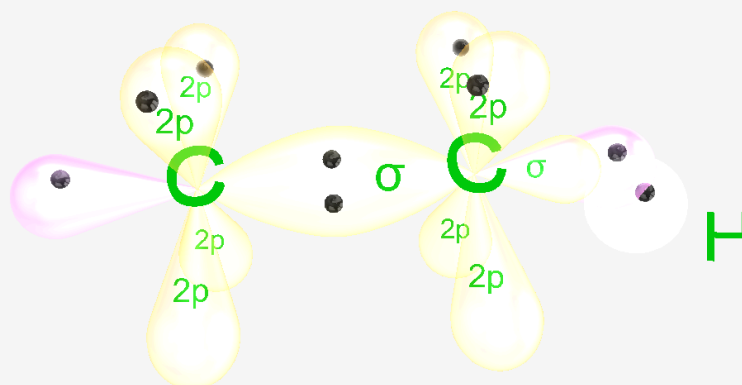
The introduction option displays the following details:

The sp^2 hybridization is used to describe the bonding in the ethene (ethylene) molecule. When a bond is formed between carbon atoms, two sp^2 orbitals overlap to form a σ -bond, while the unhybridized p-orbitals of each carbon atom overlap above and below the plane of the bonded atoms to form a single π -bond. This results in a double bond, consisting of one σ -bond and one π -bond.

When a bond is formed between a carbon atom and a hydrogen atom, the sp^2 orbital and s-orbital overlap again to form a σ -bond, resulting in a simple bond.

This animation helps users understand the hybridization of carbon, the formation of sigma (σ) and pi (π) bonds in ethene, and electron density distribution.

Electron Density Distribution of Ethyne Molecule Triple Bond



Entire model



Introduction

Share

Capture

Augmented
Reality

Language

Interactive 3. Sigma and pi bonds in ethane, ethene and ethyne.



More information for interactive 3

An interactive animation illustrates the electron density distribution of the ethyne molecule's triple bond. It displays two sp -hybridized carbon atoms in a linear geometry. Each sp carbon has two sp lobes in a straight line and two unhybridized $2p$ lobes oriented perpendicularly to each other. Each lobe contains one electron.

These two carbon atoms interact, with one sp lobe from each carbon overlapping to form a sigma bond between $C-C$.

One hydrogen atom, with an s-orbital containing one electron, interacts with the remaining sp orbital of each carbon atom, forming sigma (σ) bonds between C–H.

The two unhybridized 2p orbitals on each carbon overlap sideways, forming two pi (π) bonds, resulting in a $\text{C} \equiv \text{C}$ triple bond.

A carousel selector with arrows is at the bottom, displaying the entire model. Users can rotate the model to view it from different angles. Below it is a seek bar with a play or pause button, allowing users to demonstrate the steps of bond formation.

At the bottom, introduction, share, capture, augmented reality, and language options are available.

The introduction option displays the following details:

Sp hybridization is used to describe the bonding in the ethylene (acetylene) molecule. When a bond is formed between carbon atoms, two sp orbitals overlap to form a single σ -bond, and the remaining four non-hybrid p-orbitals overlap to form two π -bonds, resulting in a triple bond consisting of one σ -bond and two π -bonds. The π -bond electron distribution creates a cylindrical electron density around the bond axis.

When a bond is formed between a carbon atom and a hydrogen atom, the sp orbital and s-orbital overlap again to form σ -bonds (simple bonds are formed).

This animation helps users understand the hybridization of carbon, the formation of sigma and pi bonds in ethyne, and electron density distribution.

Strength of sigma and pi bonds

Consider the length and strength of single, double and triple bonds, which is covered in [section S2.2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/) about. Which bonds were strongest?

Triple bonds are the strongest bonds, followed by double bonds, and finally single bonds. But, how much stronger is a double or triple bond compared to a single bond?

Compare the bonding in two of the organic molecules from the previous simulation: ethane and ethene. They contain a carbon–carbon single and double bond respectively. Let's analyse how much energy would have to be put in to break each bond using [section 12 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet. This gives us an idea of how strong they are (**Table 2**).

Table 2. Bond strengths.

Bond	Energy (kJ mol^{-1})
C—C	346
C=C	614

The table shows that the double bonds are certainly stronger than the single bonds, but they are not twice as strong. If a carbon–carbon double bond contained two identical single bonds then the energy it should take to break it would be 692 kJ mol^{-1} ($2 \times 346 \text{ kJ mol}^{-1}$).

The difference between the single and double bond is that a double bond has a sigma and a pi bond, while the single bond only has a sigma bond. It must be that the pi bond in the double bond is weaker than the sigma bond. How do you think triple bonds would compare to single and double bonds?

Complete the following drag and drop activity by putting the statements in the correct columns for sigma or pi bonds.

Sigma bonds	Pi bonds
Weaker	
Stronger	
Head on overlap	Can occur between s and p-orbitals
Sideways overlap	Can only occur between p-orbitals
Single bonds	Make up the first bond in a double bond and in a triple bond
Not found in single bonds	Make up the second bond in a double bond and both second and third bonds in triple bonds
Bonds cannot rotate	
Bonds can rotate	

✔ Check

Interactive 2. Comparison of sigma and pi bonds and their relative properties.

If pi bonds are weaker than sigma bonds, this indicates sideways overlap must not be as effective as head-on overlap. Why is this the case?

A sigma bond has a stronger attraction between the electron pair and positive nuclei as there is only one electron-dense region on the inter-nuclear axis, located close to the nuclei (**Figure 9**).

In a pi bond the electron density is spread across two regions, above and below the inter-nuclear axis, and is found further from the positive charge of the nucleus.

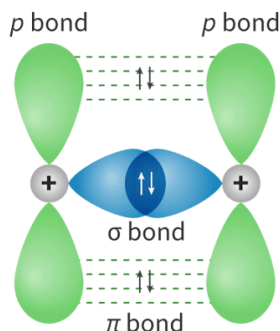


Figure 9. Sigma bonds are stronger than pi bonds.

🔗 More information for figure 9

Consider the location of these electron pairs. Which do you think would be easier to break: the sigma or the pi bond?

These two things, the location of their electrons and how significant the overlap is, form the reason that double and triple bonds are more reactive than single bonds: the pi bonds are easier to break than sigma bonds.

Study skills

Many students have a misconception that a sigma bond is the result of the overlapping of s-orbitals and a pi bond is the result of the overlapping of p-orbitals because they link the 's' to 'sigma' and the 'p' to 'pi'. This is not what sigma and pi bonds are based on.

Sigma and pi bonds are based on the way in which the orbitals overlap, not the orbitals that are overlapping. Sigma bonds can be formed by the overlapping of both the s- and p-orbitals and not just s-orbital.

Pi bonds and resonance

As part of [section S2.2.11 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/resonance-structures-hl-id-45134/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/resonance-structures-hl-id-45134/), you identified that resonance occurs in molecules and ions where a multiple bond (e.g. double bond) can be placed in more than one position in a valid Lewis formula, as in the ozone molecule in **Figure 10**.



Figure 10. Resonance structures of ozone.

🔗 More information for figure 10

The image displays two resonance structures of an ozone molecule. Each structure consists of three oxygen atoms. In the left structure, the first oxygen is single-bonded to the second oxygen, which is double-bonded to the third oxygen. In the right structure, the bonding order is reversed, with the first oxygen double-bonded to the second, which is single-bonded to the third. Double-headed arrows between the structures indicate resonance, showing how the positions of the double bonds can shift. A negative charge is indicated on the oxygen atoms with single bonds, illustrating the delocalization of electrons in resonance structures.

[Generated by AI]

The sigma bonds between each of the oxygen atoms overlap end-to-end, leaving p-orbitals to hold the electron pair in resonance above and below the bond axis. The p-orbitals overlap side-by-side to form a pi bond. In the case of ozone, three p-orbitals, one from each oxygen atom, overlaps together. In **Figure 11**, you can see how the pair of electrons held in the overlapped p-orbitals keeps the electron density above and below the axial plane.

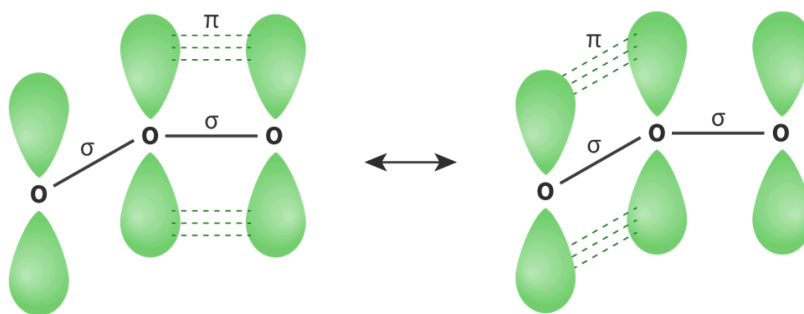


Figure 11. Representing ozone using sigma and pi notation.

 More information for figure 11

The diagram shows the molecular structure of ozone illustrating sigma (σ) and pi (π) bonds. It consists of two parts, each depicting a resonance structure of ozone. In both diagrams, three oxygen atoms are connected, represented by 'O'. Sigma bonds (σ) are indicated as solid lines connecting the O atoms directly, forming a linear arrangement. Pi bonds (π) are represented as dashed lines above and below the sigma bonds, symbolizing the delocalization of electron density across the oxygen atoms. The left structure shows two pi bonds on one side, with sigma bonds holding the entire structure together. The right structure appears as a mirror image of the left, with one pi bond on the opposite side of the central sigma bond, highlighting the resonance and electron delocalization in ozone molecules. Each p-orbital in the structure extends outward, representing the potential space where the electron density can be found.

[Generated by AI]

To form this resonance hybrid structure (**Figure 12**), are the electrons that become delocalised located in a sigma or a pi bond?

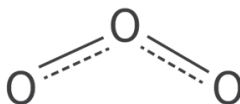


Figure 12. Resonance hybrid structure of ozone.

 More information for figure 12

The image is a diagram depicting the resonance hybrid structure of ozone (O_3), consisting of three oxygen atoms. It shows three circles representing the oxygen atoms arranged in a V-shape. The central oxygen atom is connected to the two outer oxygen atoms by dashed lines, indicating resonance bonds that are shared between pi bonds. These lines illustrate the delocalization of pi electrons within the molecule. There are no specific values or labels apart from the atoms and resonance lines.

[Generated by AI]

If it is the second bond in a double bond, then it must be the pi electrons that have become delocalised.

Nature of Science

Aspect: Theories

The formation of new hybrid molecular orbitals is a key point of molecular orbital theory. It explains the bonding in a number of molecules, including those who violate the octet rule and those that form coordination bonds. It also provides a model for describing the energies and probable location of the electrons involved in bonds.

While a deep understanding of this theory is not required at this level, this is a theory that you may learn a lot about if you continue to study chemistry at university level.

In the activity in [section S2.2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/), you may remember that you used your fingers to represent differences in single, double and triple bonds. The first part of this activity will require you to do something similar to help you consolidate your understanding of the bond rotation associated with single and double bonds, but this time you will be using pencils and paper.

The second part of the activity allows you to compare these two types of bond in their structure and across a range of properties. You can choose to communicate your understanding in the second part using either a table or a Venn diagram.

Activity

- IB learner profile attribute: Inquirer

- **Approaches to learning:** Thinking skills — Designing procedures and models
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

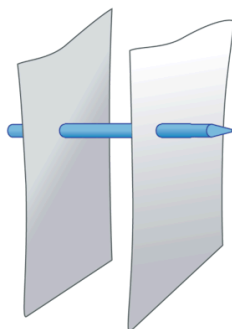
Part 1: Bond rotation

Materials

- Pencil × 2
- Sharpener
- Two pieces of paper

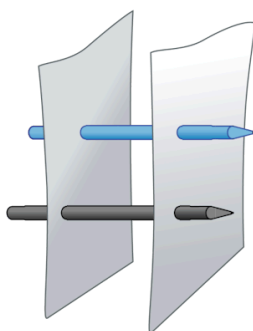
Method

1. Take a pencil and push it through two sheets of paper, spacing the two sheets far apart, as you can see in the diagram below. Try to rotate the papers.



 More information

2. Sharpen the second pencil so it is at least 2 cm smaller than the first pencil that you used in Step 1.
3. Push the second pencil through the same sheets of paper you used in Step 1, placing it just below the first pencil. Space the papers as part apart as possible. Try to rotate the papers.



Questions

1. Compare the distance between the papers in the first step and the third step. Suggest what this can be used to indicate about bond length.
2. Outline how Step 1 can be used to represent the ability of a single bond to rotate. Compare this to Step 3.
3. Suggest why it was important to sharpen the second pencil.

Part 2: Sigma and pi comparison

Create a table or Venn diagram to compare sigma and pi bonds. Include at least **five** headings and two diagrams for each type of bond.